Rajshahi University of Engineering & Technology

Department Of Mechatronics Engineering

AI-Enhanced Stereo Matching for High-Accuracy Depth Mapping and 3D Reconstruction

Course No: MTE 4200

Course Title: Project and Thesis

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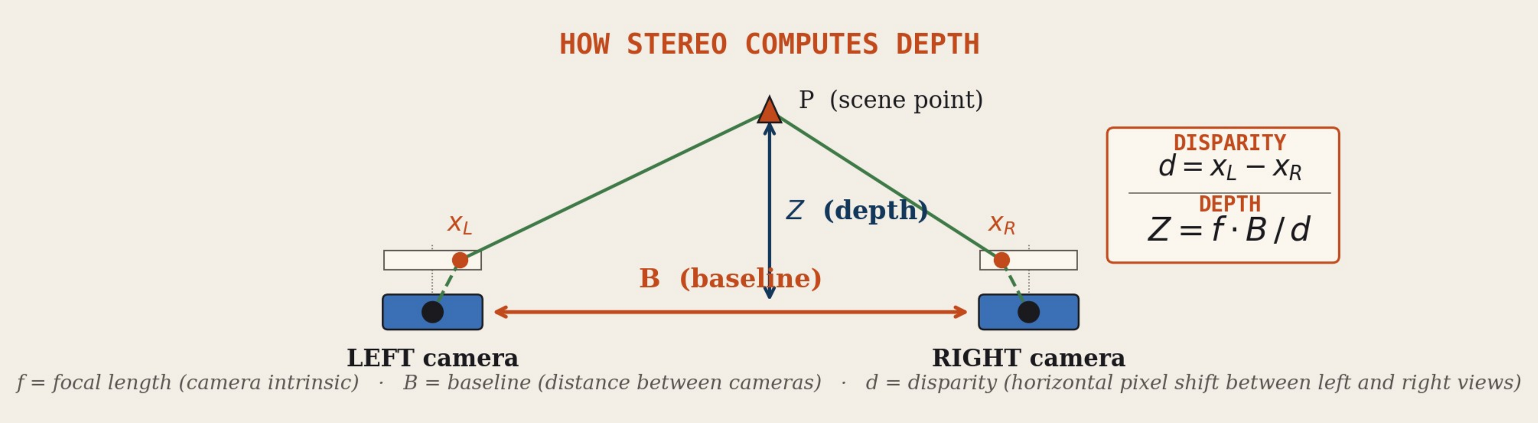
Outline

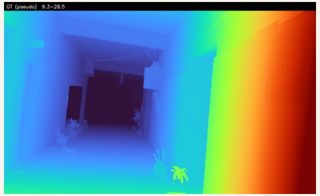
Introduction

Stereo cameras turn a horizontal pixel shift into metric depth.



REAL EXAMPLE · indoor hallway from our finetune set

Left view



Depth map

Introduction (Cont..)

We build StereoLite, a 2.96 M parameter stereo network designed to run on edge hardware.

Every moving platform that must understand its surroundings needs depth, in real time, on board. The same model has to fit and run on whatever compute the platform can carry.

WHERE ON-DEVICE STEREO DEPTH MATTERS

Drones

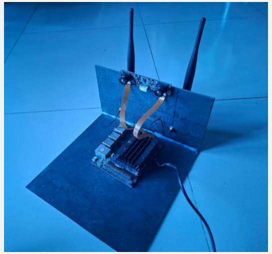
obstacle avoidance, terrain following

Mobile robots

navigation, picking, collision safety

AR / VR

scene reconstruction, hand-held depth



Embedded rigs

SLAM, mapping, autonomous platforms

Problem Statement

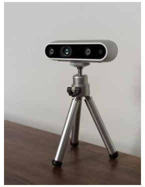
How do we deliver dense, accurate depth on edge hardware, when the obvious alternatives don't fit?

WHY NOT THE OBVIOUS ALTERNATIVES?

LiDAR

$3 k to $80 k

sparse, heavy, power-hungry; cost rules out consumer platforms



Active depth (RealSense)

≤ 6 m, indoor

IR projector fails outdoors; range and resolution limited

~340 M params

Foundation stereo

state-of-the-art accuracy, but needs a desktop GPU; can't ship to a Jetson

EDGE CONSTRAINTS WE TARGET

LIMITED COMPUTE

~6 TOPS

Embedded SoCs deliver a fraction of a desktop GPU.

TIGHT MEMORY

~4 GB

Shared with the rest of the autonomy stack.

POWER BUDGET

5 to 25 W

Battery-powered platforms, no hot GPUs allowed.

Objectives

Two Clear Goals

1. To design a computationally efficient stereo matching pipeline, leveraging AI-based disparity refinement to enhance depth estimation on resource-limited platforms.
2. To design an architecture able to withstand camera rectification imperfections.

To Achieve This

PARAMETERS

< 5 M

INFERENCE

< 60 ms

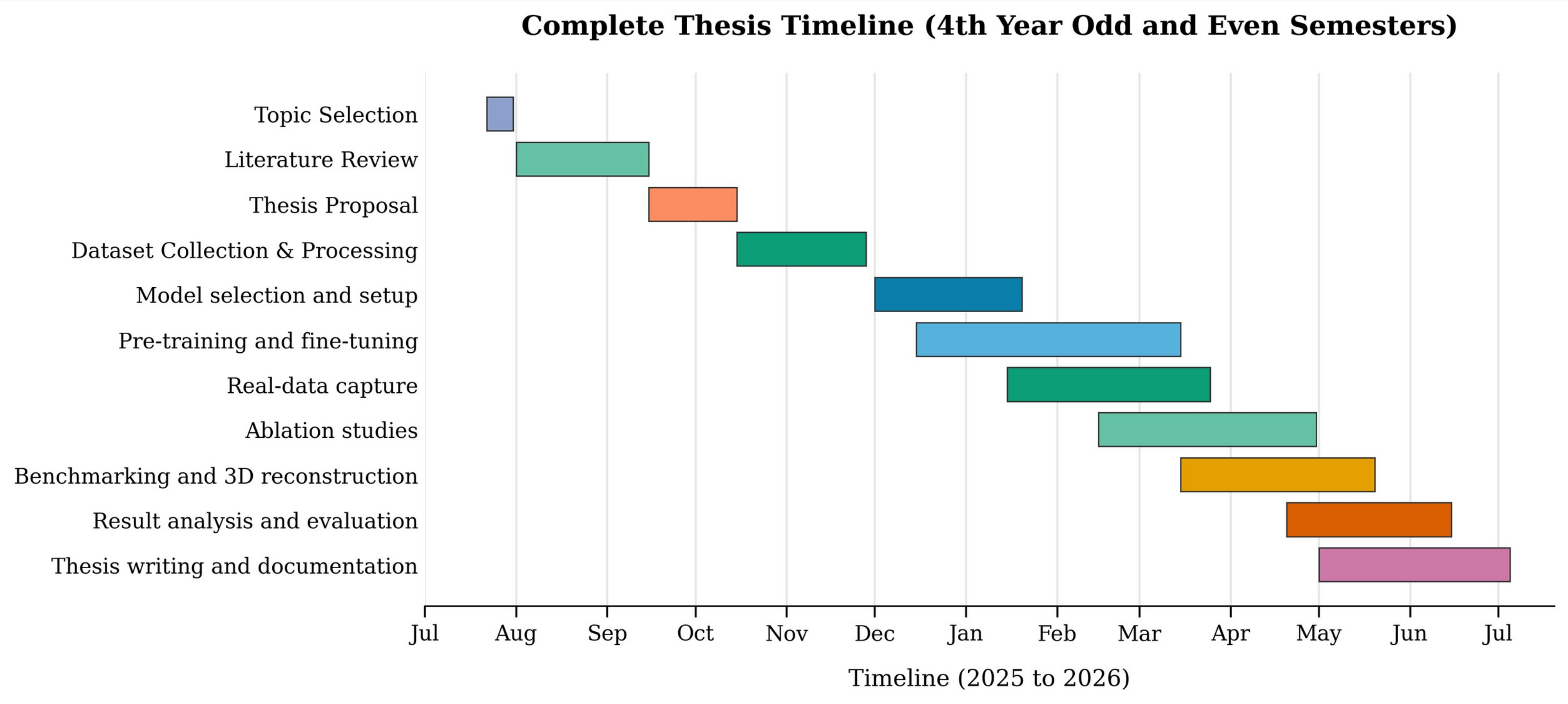
Jetson Orin Nano

CHECKPOINT

< 20 MB

Architecture must work on cheap stereo cameras with imperfect alignment and missing rectification, end-to-end trainable, multi-scale supervision so the network **absorbs small calibration errors**.

Time Plan



Eleven overlapping phases, July 2025 to July 2026 (4th year odd and even semesters)

Literature Review

Comparison of nine prior methods plus ours on the standard Scene Flow + KITTI 2015 protocol

#	Method	Year	Type	Params (M)	SF EPE (px)	KITTI D1 (%)	Latency	Edge?
1	PSMNet	2018	3D cost vol.	5.2	1.09	2.32	410 ms	No
2	HITNet (L)	2021	Tile-based	0.97	0.43	1.98	54 ms	Yes
3	BGNet	2021	Bilateral grid	2.9	1.17	2.51	25 ms	Yes
4	CoEx	2021	Lightweight	2.7	0.69	2.13	27 ms	Yes
5	RAFT-Stereo	2021	Iterative	11.2	0.61	1.82	380 ms	No
6	IGEV-Stereo	2023	Iter. + GEV	12.6	0.47	1.59	180 ms	No
7	LightStereo-S	2025	Lightweight	3.4	0.73	2.30	17 ms	Yes
8	FoundationStereo	2025	Foundation	~340	0.34	—	—	No
9	DEFOM-Stereo	2025	Foundation	47.3	0.42	—	316 ms	No
*****	**StereoLite (Ours)**	**2026**	**Tile + Iter.**	**2.96**	**0.78**	**3.93***	**50 ms**	**Yes**

Latencies on different GPUs (varies by source); SF EPE on Scene Flow finalpass; KITTI 2015 D1-all from official leaderboards, where each method is fine-tuned on KITTI.
** Ours is ZERO-SHOT on the KITTI 2015 training split under our protocol (no KITTI fine-tuning). See the References slide for citations.*

Literature Review (Cont..)

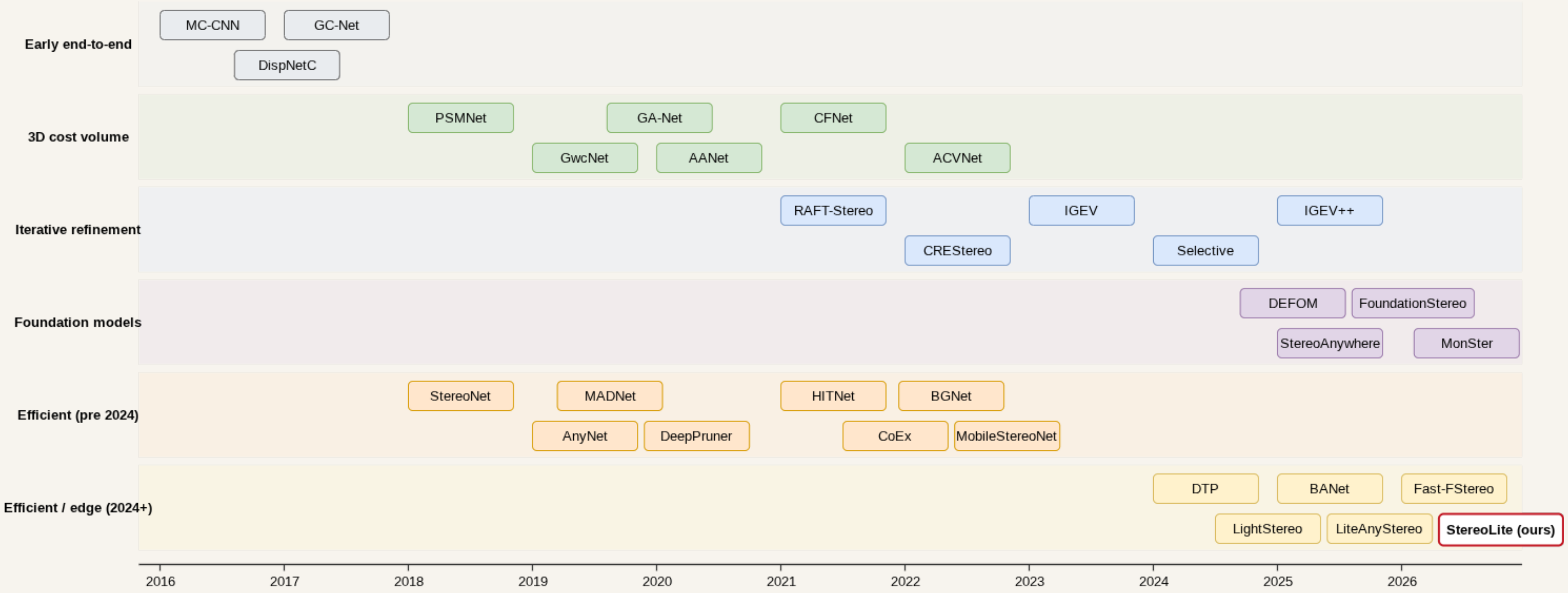
Capability matrix · what each method offers for edge stereo

Method	≤3 M params	<60 ms edge	Iterative	Plane / tile	Foundation	Cross-domain
PSMNet [1]	✗	✗	✗	✗	✗	✗
HITNet (L) [2]	√	√	✗	√	✗	✗
BGNet [3]	√	√	✗	✗	✗	✗
CoEx [4]	√	√	✗	✗	✗	✗
RAFT-Stereo [5]	✗	✗	√	✗	✗	√
IGEV-Stereo [6]	✗	✗	√	✗	✗	√
LightStereo-S [7]	~	√	✗	✗	✗	✗
FoundationStereo [8]	✗	✗	√	✗	√	√
DEFOM-Stereo [9]	✗	✗	√	✗	√	√
StereoLite (Ours)	√	√	√	√	✗	√

StereoLite is the only method combining lightweight, real-time, iterative, and plane-tile geometry simultaneously, and it generalizes zero-shot across four unseen datasets (Middlebury still trails foundation-prior methods).

Evolution of Stereo Matching

Six visible eras, classical to foundation-model

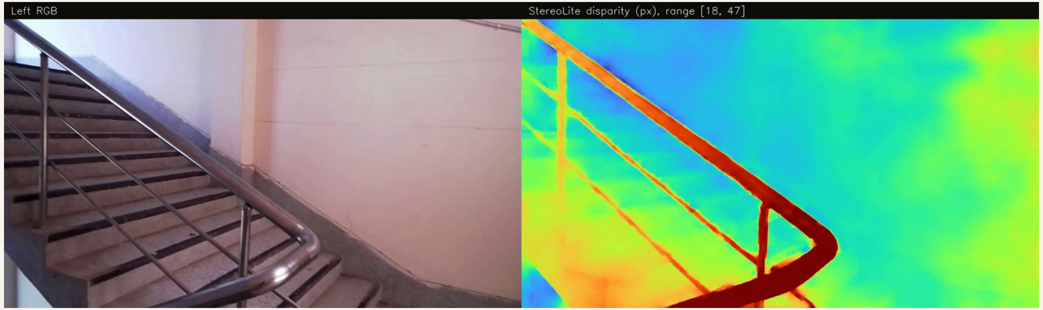


Timeline of stereo matching methods, 2002 to 2026

Edge-oriented designs form their own track from 2018 onward; StereoLite builds on that track.

Proposed Solution

Introducing StereoLite



Indoor inference · 10 s clip · 1.5x slowdown

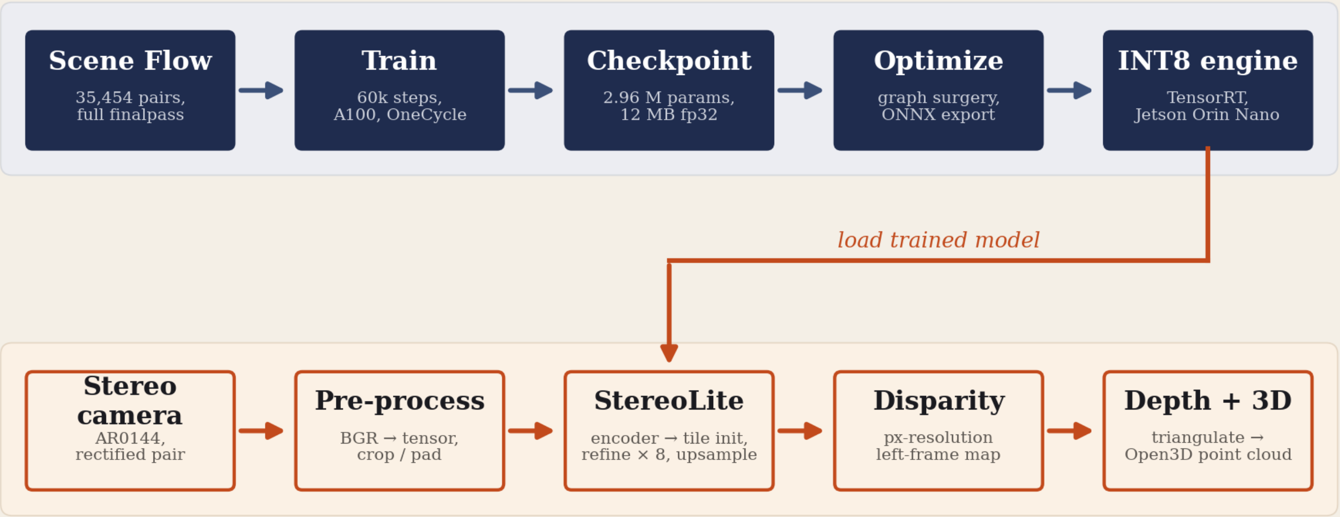
3D reconstruction from disparity

PARAMETERS	LATENCY · ORIN NANO	EPE · SYNTHETIC	D1 · ZERO-SHOT
2.96M	**36.3ms**	**0.78px**	**10.9%**
12 MB checkpoint (fp32)	INT8 on Jetson Orin Nano, 384x640	EPE on Scene Flow FT3D test, 4,370 pairs	zero-shot Middlebury 2014 D1-all

An edge sized stereo network that learns dense depth from two cheap cameras, end to end.

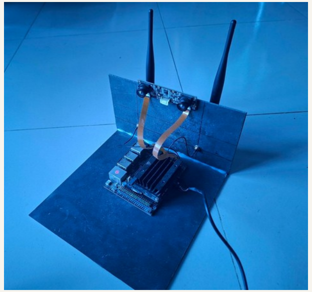
Methodology

System overview · training pipeline (top) and inference pipeline (bottom)



Implementation

HARDWARE



Test rig

Stereo camera

Jetson Orin Nano

SOFTWARE

CUDA

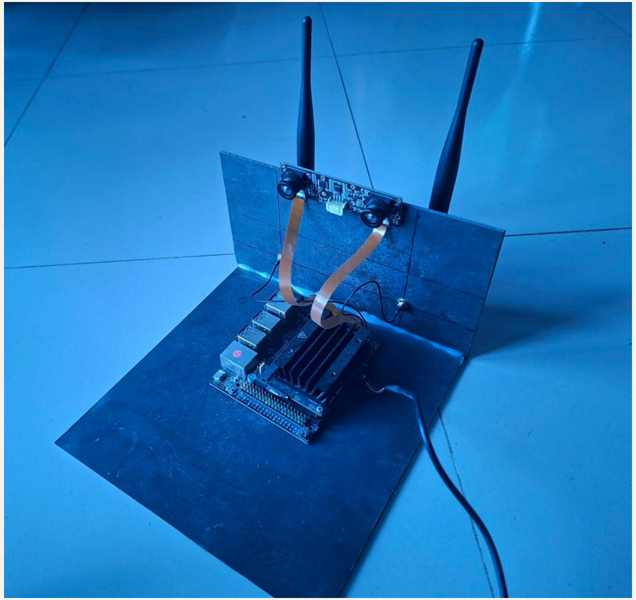
PyTorch

Open3D

Kaggle

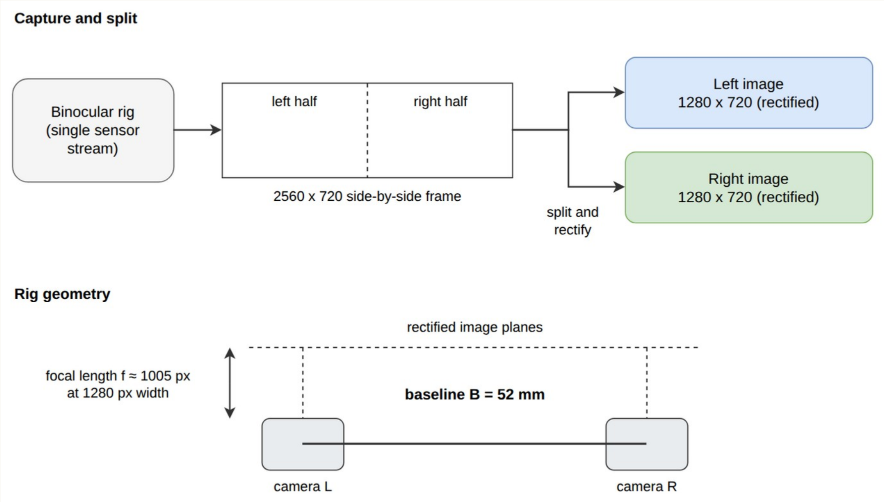
Experimental Setup

Low-cost stereo rig and rig geometry



Stereo test rig

AR0144 stereo camera, 2560 x 720 side-by-side capture.

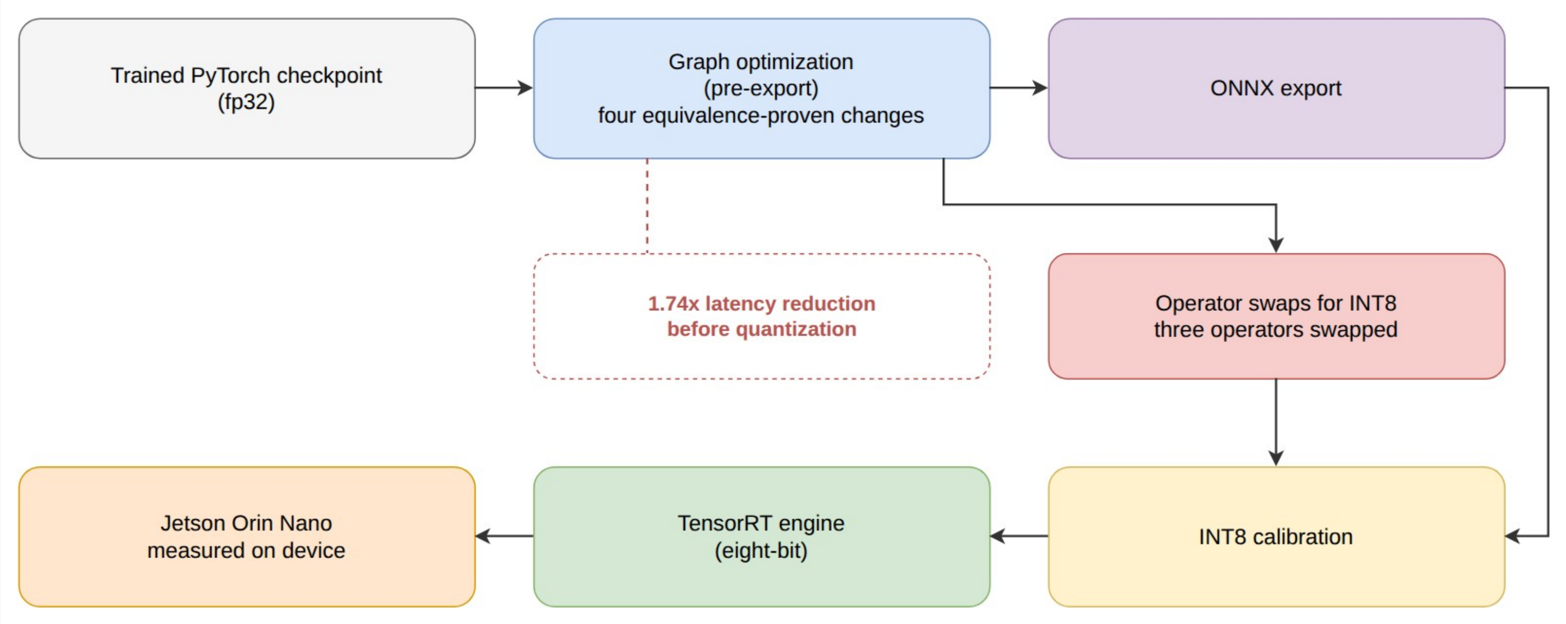


Rig geometry

Baseline 52 mm, focal length about 1005 px after rectification to 1280 x 720.

Deployment Pipeline

From fp32 checkpoint to a Jetson INT8 engine

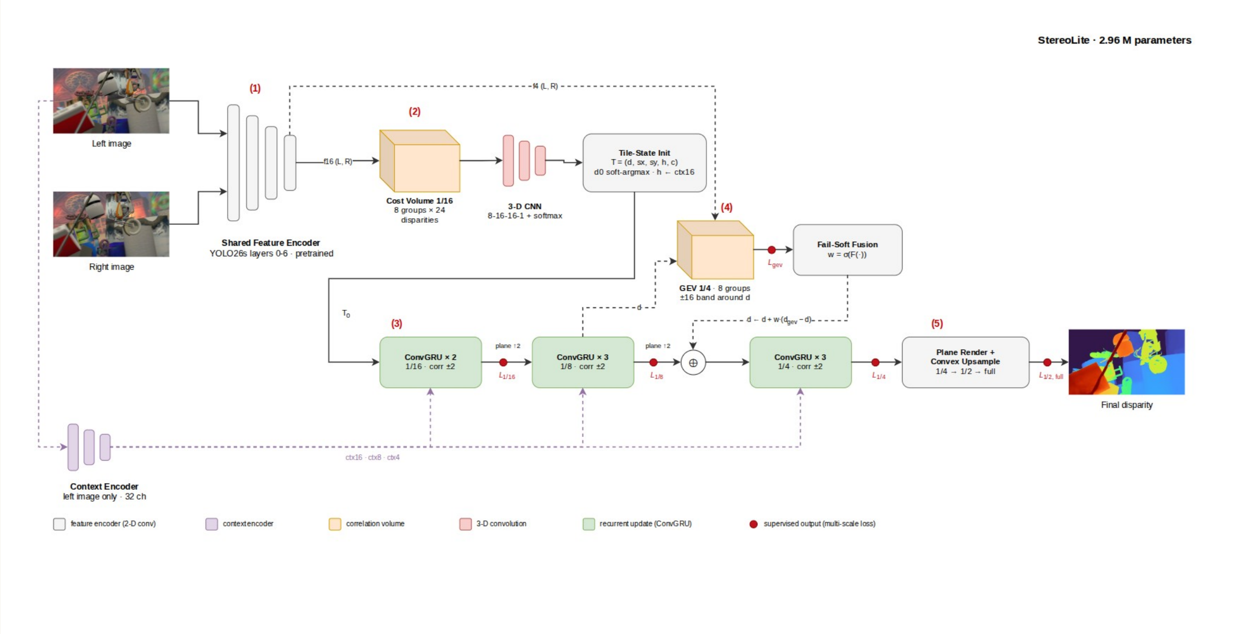


Export path: graph optimization, ONNX, INT8 calibration, TensorRT

Four equivalence-proven graph changes give a 1.74x latency cut before quantization; three operator swaps make the graph INT8-friendly.

Working Principle

StereoLite, end-to-end

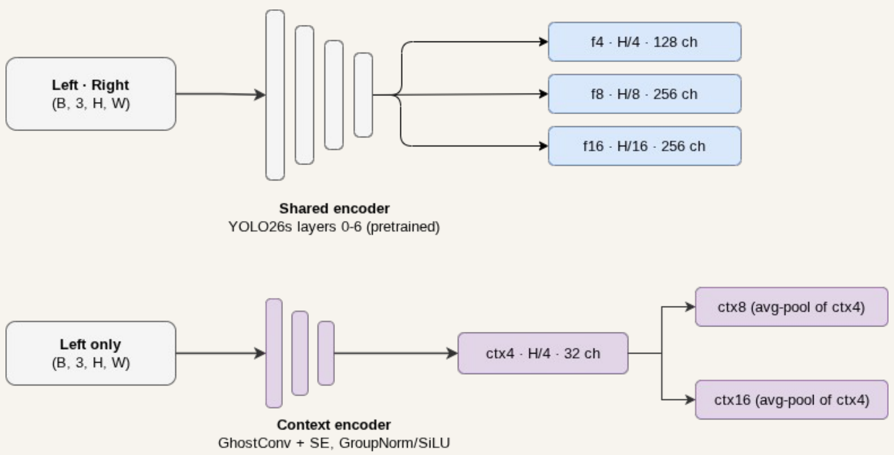


StereoLite final architecture: encoder, cost volume init, recurrent refinement with GEV fusion, plane upsampling

2.96 M trainable parameters; encoder and refinement hold roughly 95 percent of the budget

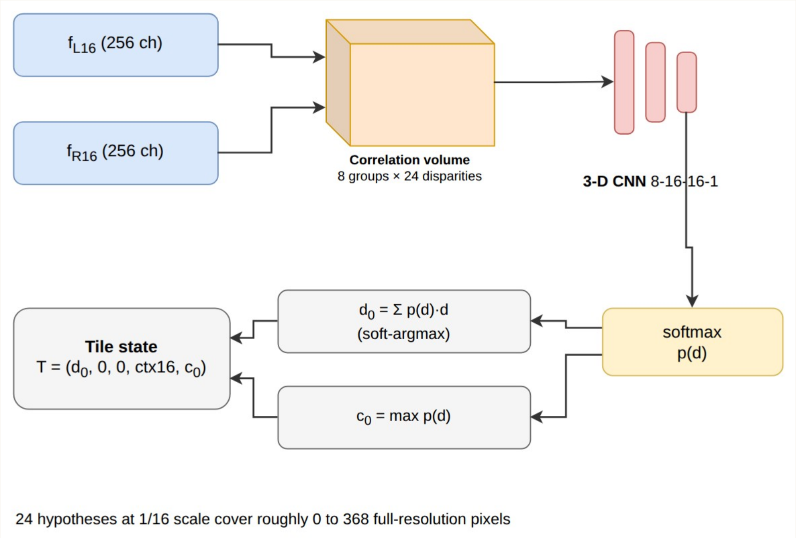
Working Principle · Stage 1 + 2

Encoders and tile hypothesis init



Stage 1 · YOLO26s encoder + context encoder

A truncated YOLO26s backbone extracts shared features at 1/4, 1/8 and 1/16; a separate 32-channel context encoder reads the left image for the refinement stream.

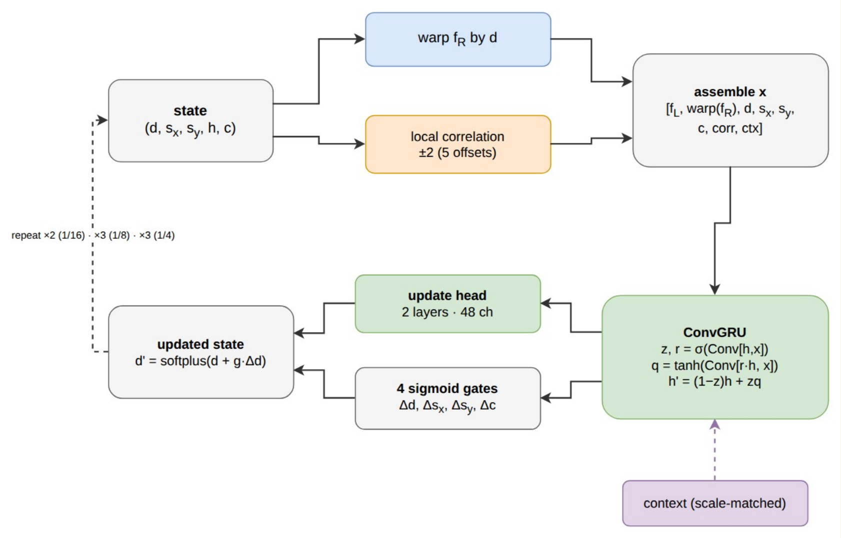


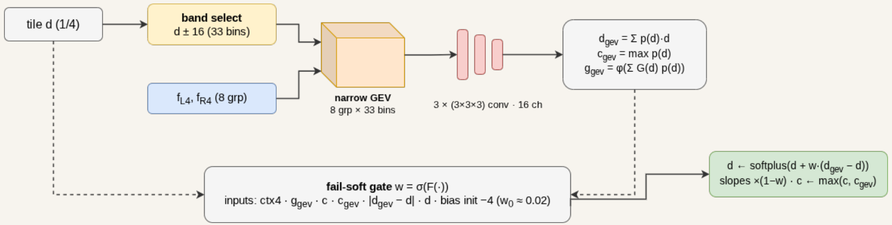
Stage 2 · group-wise cost volume + tile init

An 8-group correlation volume with 24 hypotheses at 1/16 is aggregated by a small 3D CNN; soft-argmin seeds one disparity hypothesis per tile.

Working Principle · Stage 3

Recurrent refinement with geometry evidence





Stage 3 · ConvGRU refinement, 2 + 3 + 3 = 8 updates

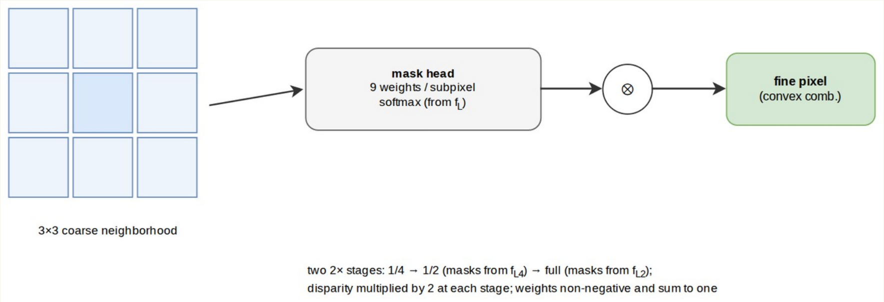
A ConvGRU refines tile disparities across 1/16, 1/8 and 1/4 using local correlation lookups and plane-aware cross-scale propagation.

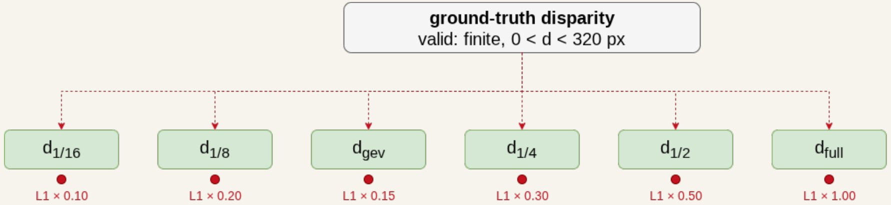
Narrow-band GEV at 1/4 · fail-soft gate

A 33-hypothesis geometry encoding volume around the current estimate adds fresh 1/4-scale evidence, fused through a learned fail-soft gate.

Working Principle · Stage 4 + Supervision

Plane upsampling and multi-scale loss





Stage 4 · plane rendering + convex upsample

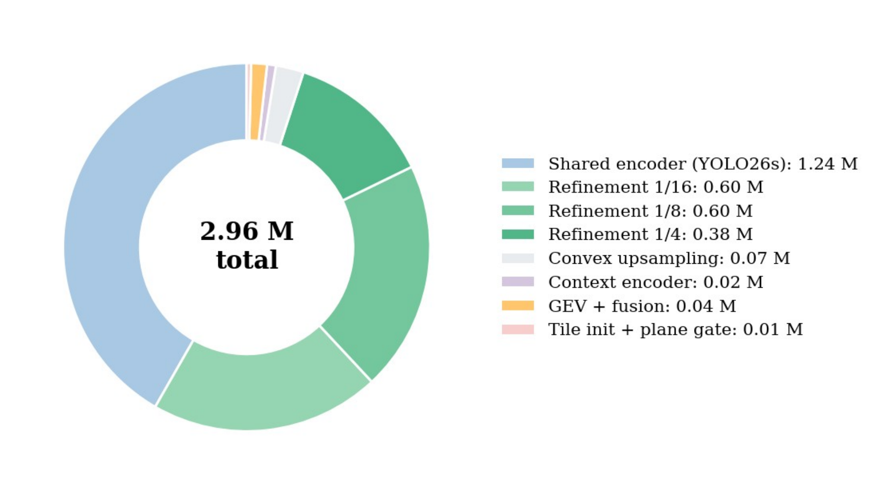
Tile planes are rendered with an edge gate, then two learned convex 2x stages restore full-resolution disparity.

Supervision · 11-term multi-scale objective

L1 at five scales plus GEV, gradient, threshold-stack, D1 hinge, smoothness and slant terms supervise every stage.

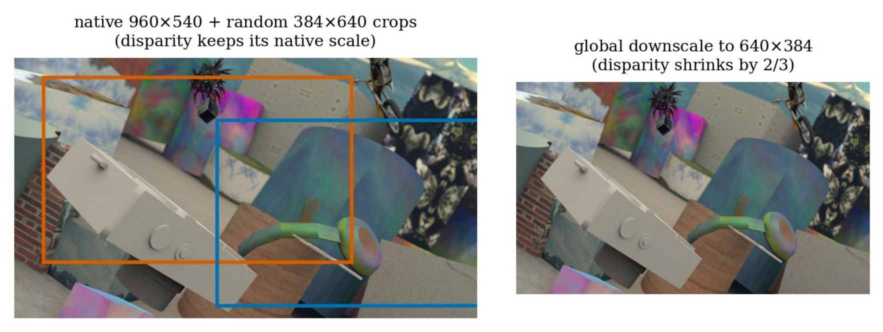
Working Principle · Budget + Input

Where the 2.96 M parameters go, and what the network sees



Module-wise parameter split

Encoder and refinement hold roughly 95 percent of the 2.96 M budget.

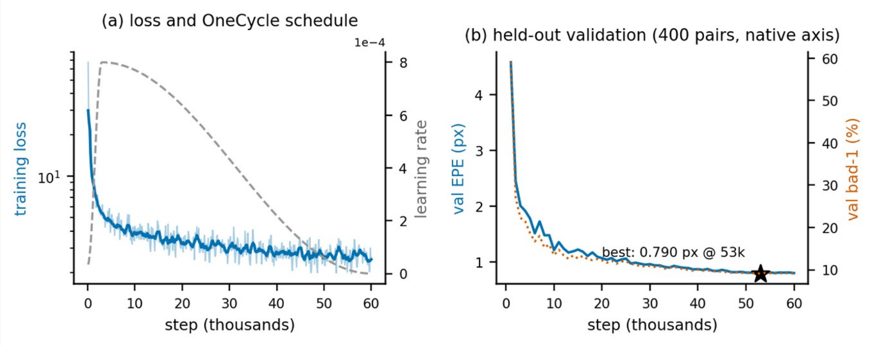


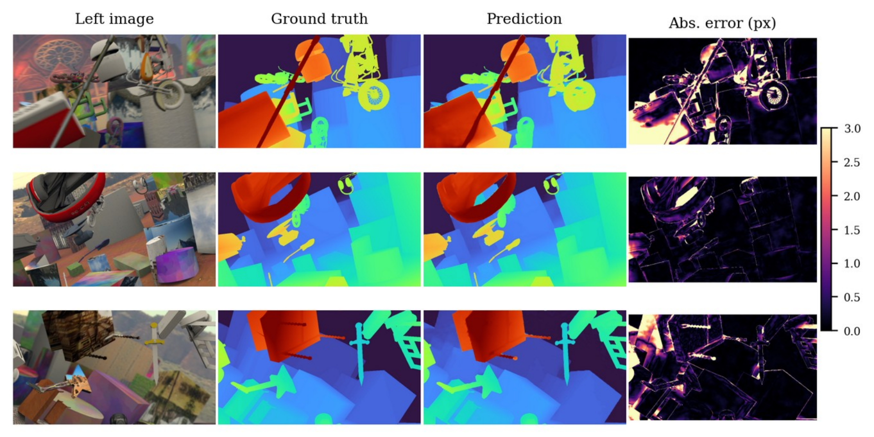
Input protocol: native 384 x 640 crops

Random co-located crops with asymmetric color jitter and right-only erase.

Results

Scene Flow pre-training, in-domain test





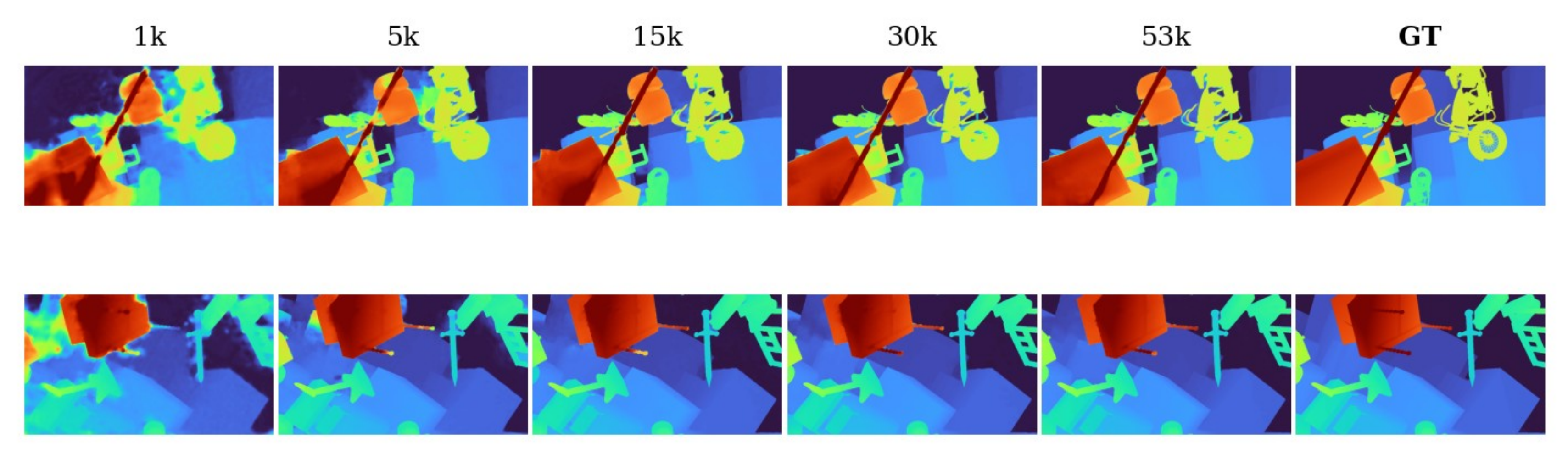
Training loss and validation EPE, 60k steps on full Scene Flow

FT3D test qualitative: left image, ground truth, prediction, error

TRAINING PAIRS	TEST EPE	D1-ALL
35,454	**0.78 px**	**3.40%**
full Scene Flow finalpass	*FT3D test set, 4,370 pairs*	*outlier rate, FT3D test*

Results: Convergence

Prediction quality across training checkpoints

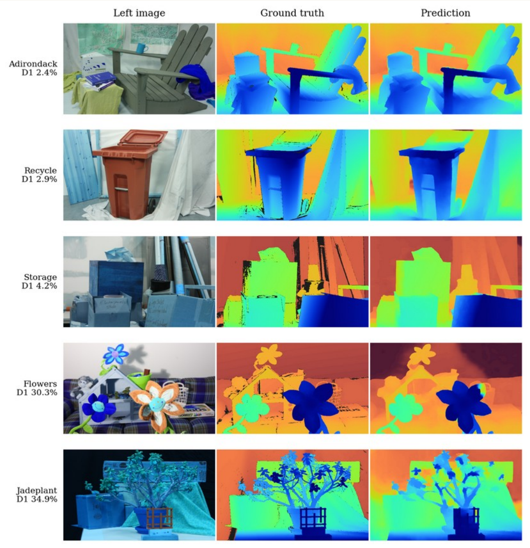


Predicted disparity at increasing training steps, two FT3D scenes

Structure appears early; thin details and boundaries keep sharpening to step 53k.

Results

Zero-shot generalization and on-device deployment



Middlebury 2014 zero-shot: easiest and hardest scenes

Zero-shot transfer to our low-cost real rig

ZERO-SHOT MB14

10.9% D1

23 scenes, no fine-tuning

REAL RIG

1.45 px

EPE vs FoundationStereo, 997 pairs

JETSON ORIN NANO

36.3 ms

27.5 FPS, INT8 TensorRT, measured

Results: Zero-Shot Quartet

One checkpoint, four unseen datasets, one protocol (384 x 640, valid disparities up to 192 px)

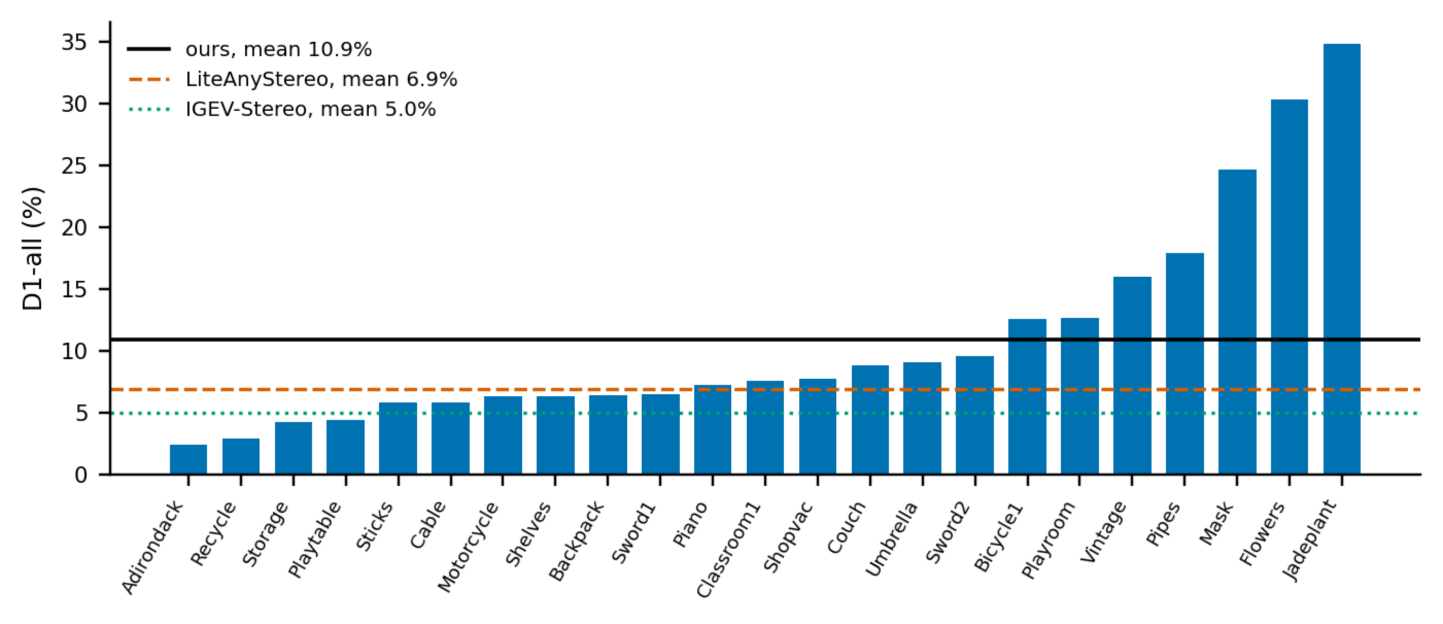
Dataset	Pairs	EPE (px)	bad-2 (%)	D1-all (%)
KITTI 2012 (train)	194	0.82	6.96	**4.33**
KITTI 2015 (train)	200	0.82	6.42	**3.93**
ETH3D (train)	27	0.93	6.47	**3.96**
Middlebury 2014	23	1.71	14.5	**10.9**
Scene Flow FT3D test (in-domain)	**4,370**	**0.78**	**5.34**	**3.40**

Driving and outdoor domains land near the in-domain outlier rate; indoor close-range (Middlebury) remains the weak axis.

Training-split evaluations under our protocol; official leaderboard submissions pending.

Results: Zero-Shot Per Scene

Middlebury 2014, all 23 perfect-set scenes

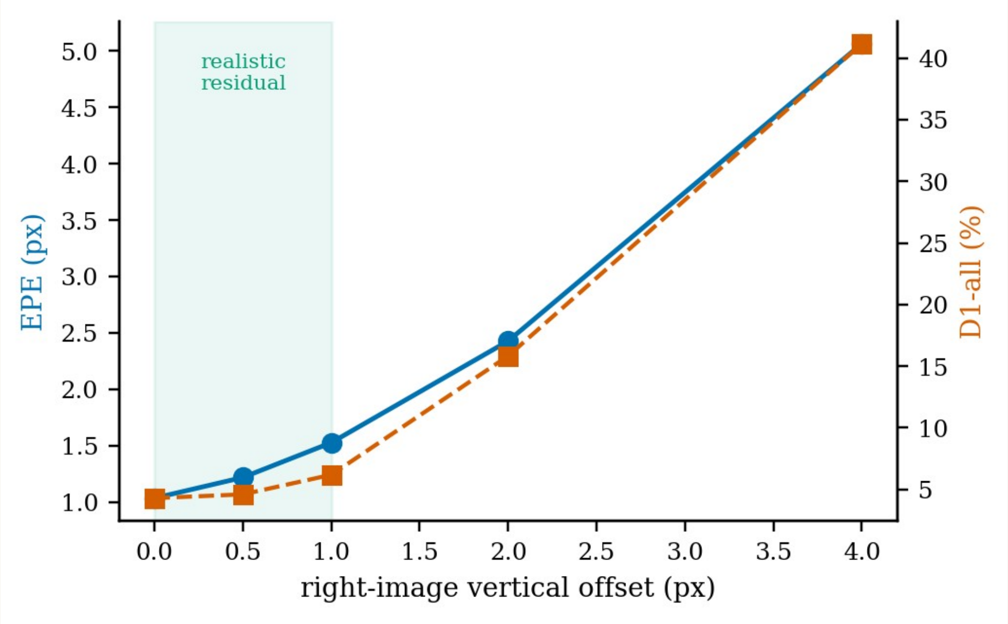


Per-scene D1-all, sorted

16 of 23 scenes land in the reference band; Jadeplant and Flowers are the hardest.

Results: Rectification Robustness

Accuracy under vertical misalignment (second objective)

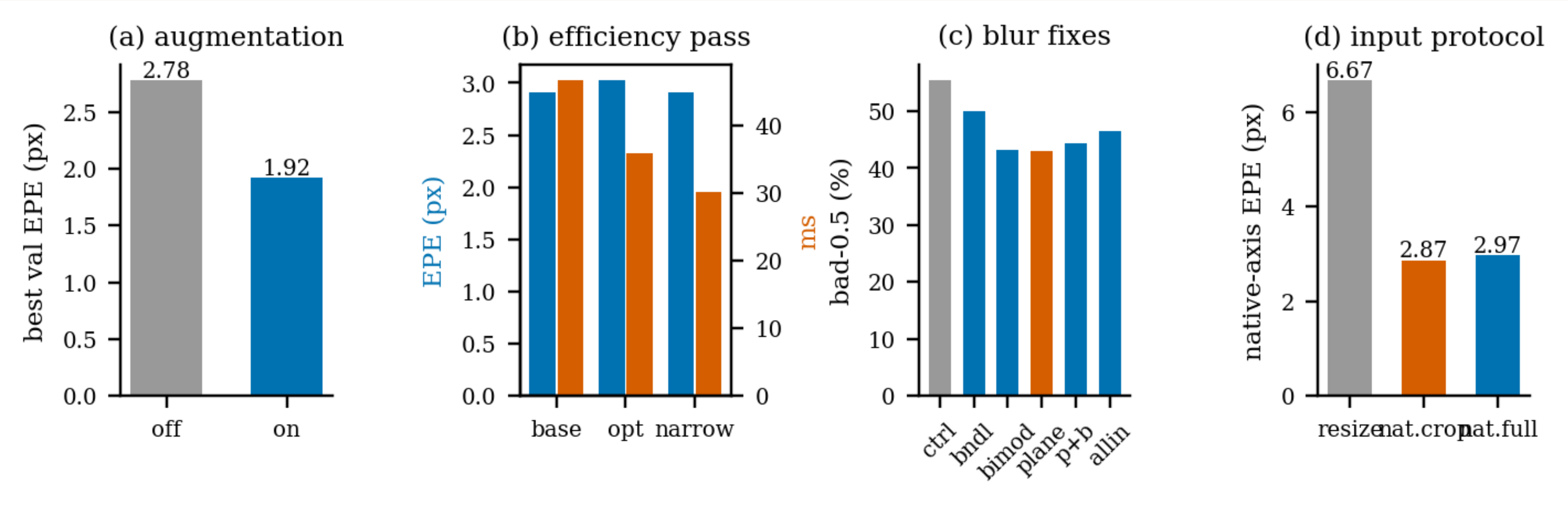


EPE and D1 vs injected vertical offset

EPE rises only 1.03 to 1.53 px up to 1 px of offset; degradation becomes severe beyond 2 px.

Results: Ablations

What each design and training choice contributes

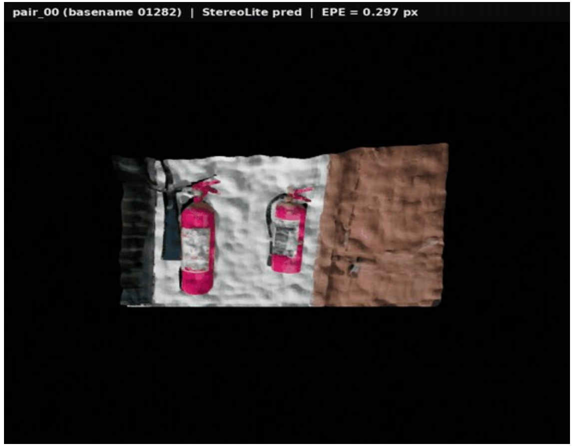


Augmentation, inference optimization, blur and input protocol ablations

Every retained choice is backed by a controlled A/B on the same pair set.

Results

3D point cloud reconstruction



val pair · EPE 0.297 px

val pair · EPE 0.258 px

val pair · EPE 0.315 px

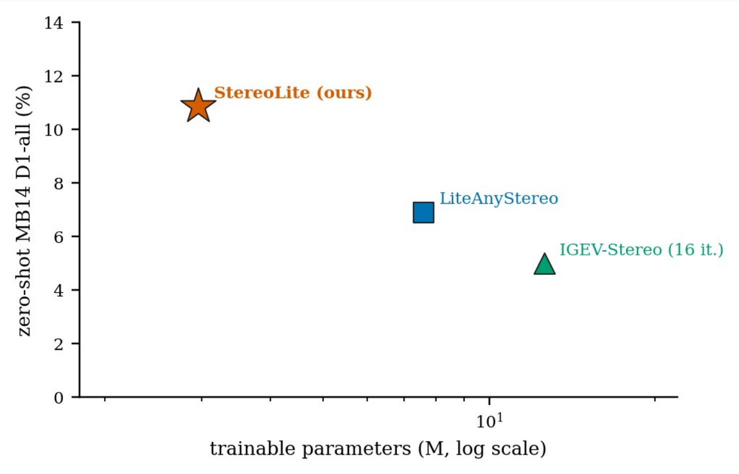
CAMERA INTRINSICS

AR0144: $f_x \approx 1005$ px, baseline 52 mm. Pinhole projection · Open3D post-processing.

Confirms the predicted disparity is **metrically consistent** , not just visually plausible.

Discussion

- In-domain: 0.78 px EPE, 3.40% D1 on the full FT3D test set at 2.96 M parameters.
- Zero-shot: D1 4.33% (KITTI 2012), 3.93% (KITTI 2015), 3.96% (ETH3D); driving and outdoor domains land near the in-domain outlier rate.
- Middlebury 2014 is the remaining weak axis: 10.9% D1, within 4 points of LiteAnyStereo at 2.6x fewer parameters.
- Robust: EPE rises only 1.03 to 1.53 px up to 1 px of vertical misalignment.
- Deployable: 36.3 ms (27.5 FPS) INT8 on Jetson Orin Nano, measured on device.
- Real rig: 1.45 px mean EPE agreement with the FoundationStereo teacher over 997 pairs.



Zero-shot accuracy vs parameters

Challenges / Limitations

- Hardest Middlebury scenes stay weak: Jadeplant 34.9% D1, Flowers 30.3%, both textureless and repetitive structure.
- Accuracy degrades beyond 2 px vertical misalignment: D1 15.8% at 2 px, rising to 41.2% at 4 px.
- Recurrent reasoning stops at quarter resolution, limiting recovery of thin and distant structures.
- Supervision is synthetic only, and results come from a single training run.
- KITTI and ETH3D measured on training splits under our own protocol; official leaderboard submissions still pending.
- Real-rig accuracy is agreement with the FoundationStereo teacher, not absolute ground truth.

Impact

Original work & deployment

KEY CONTRIBUTION

First published combination of HITNet tile hypotheses + ConvGRU iterative refinement + a narrow-band GEV at 2.96 M parameters . Original work, not a re-implementation.

WHERE IT HELPS

Drones

Indoor & dense forest obstacle avoidance

AGVs

Factory navigation without LiDAR mast

AR / VR

On-device hand & scene tracking

SLAM

Visual SLAM & ADAS on Jetson SoCs

Plus: an honest, open baseline for the sub-3 M parameter edge stereo regime.

Conclusion

What we built · what we measured · what we set out to do

STEREOLITE

A 2.96 M parameter stereo network: tile plane hypotheses with ConvGRU iterative refinement, trained on the full Scene Flow set (35,454 pairs). FT3D test EPE 0.78 px / D1-all 3.40%. Zero-shot Middlebury 2014 D1 10.9%, within 4 points of LiteAnyStereo at 2.6x fewer params. Measured on device: 36.3 ms INT8 (27.5 FPS) on Jetson Orin Nano; 49.8 ms fp16 on RTX 3050.

Objective	Stated criterion	Status
Computationally efficient pipeline	Real-time on Jetson Orin Nano	**Met · 36.3 ms INT8 (measured)**
Accurate disparity estimation	Competitive on Scene Flow test	**Met · EPE 0.78 px, D1 3.40%**
Cross-domain generalization	Zero-shot on unseen datasets	**Met · KITTI 3.9/4.3%, ETH3D 4.0%, MB14 10.9%**
Camera-imperfection tolerance	Works on a real, imperfect rig	**Met · 1.45 px on 997 real pairs**

Robust to rectification error (EPE 1.03 to 1.53 px up to 1 px vertical offset); all stated objectives are met.

Future Work

1. Three-stage knowledge distillation: synthetic supervision, then self-distillation under input perturbation, then a frozen foundation teacher on unlabeled real pairs.
2. Submit to the official KITTI and ETH3D leaderboards; our training-split zero-shot results already sit at D1 4.33, 3.93 and 3.96 percent.
3. Deployment hardening: full accuracy audit of the INT8 engine against the fp32 model, then ports to other edge boards.
4. Temporal extension: propagate tile-plane state across video frames to amortize initialization cost.
5. Mapping extension: fuse per-frame point clouds into a persistent scene model.

Impact (Cont..)

Environmental · Societal · Research

RESEARCH

- Combines tile plane hypotheses (HITNet) with ConvGRU iterative refinement (RAFT) and a narrow-band GEV in one 2.96 M model.
- Teacher-verified on real data: 1.45 px agreement with FoundationStereo over 997 rig pairs.
- Open, reproducible baseline for edge stereo under 3 M parameters.

ENVIRONMENTAL

- ~30× lower inference power vs running a 340 M foundation stereo model on a server GPU (projected, edge GPU at 7 to 15 W).
- No cloud roundtrip per depth frame: all inference local.
- Enables battery-powered stereo on drones, AR headsets, mobile robots.

SOCIETAL

- On-device inference: images never leave the device.
- ~USD 500 stereo + edge GPU setup, vs USD 2 k+ for comparable LiDAR depth.
- Lowers the entry bar for stereo research and education in resource-constrained settings.

References

IEEE-style citations of methods featured in the literature review

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[2] V. Tankovich, C. Häne, Y. Zhang, A. Kowdle, S. Fanello, and S. Bouaziz, "HITNet: Hierarchical iterative tile refinement network for real-time stereo matching," in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2021, pp. 14362–14372.

[3] B. Xu, Y. Xu, X. Yang, W. Jia, and Y. Guo, "Bilateral grid learning for stereo matching networks," in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2021, pp. 12497–12506.

[4] A. Bangunharcana, J. W. Cho, S. Lee, I. S. Kweon, K.-S. Kim, and S. Kim, "Correlate-and-Excite: Real-time stereo matching via guided cost volume excitation," in Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS), 2021, pp. 3542–3548.

[5] L. Lipson, Z. Teed, and J. Deng, "RAFT-Stereo: Multilevel recurrent field transforms for stereo matching," in Proc. Int. Conf. 3D Vision (3DV), 2021, pp. 218–227.

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Q & A

Thank You

Code, training logs, and curves available on request.

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